


```
import torch

print("CUDA Available:", torch.cuda.is_available())

if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))
```

CUDA Available: True
GPU: Tesla T4

```
!pip install diffusers transformers accelerate safetensors
```

Requirement already satisfied: diffusers in /usr/local/lib/python3.12/c
Requirement already satisfied: transformers in /usr/local/lib/python3.1
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/
Requirement already satisfied: safetensors in /usr/local/lib/python3.12/
Requirement already satisfied: importlib-metadata in /usr/local/lib/pyt
Requirement already satisfied: filelock in /usr/local/lib/python3.12/di
Requirement already satisfied: httpx<1.0.0,>=0.34.0 in /usr/loc
Requirement already satisfied: huggingface-hub<2.0,>=0.34.0 in /usr/loc
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/pyth
Requirement already satisfied: requests in /usr/local/lib/python3.12/di
Requirement already satisfied: Pillow in /usr/local/lib/python3.12/dist
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/pythor
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/loc
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.1
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dis
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/c
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/pytho
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/p
Requirement already satisfied: typer>=0.20.0 in /usr/local/lib/python3.
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/l
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/pythor
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/p
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/loc
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/lo
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/l
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/l
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.0.0.* in

```

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/
Requirement already satisfied: nvidia-cuspars-cu12==12.5.8.93.* in /usr/
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/
Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.12/
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/pyt
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/p
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/pyt
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.1
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/pyt
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.1

```

```

from diffusers import StableDiffusionPipeline
import torch

```

Flax classes are deprecated and will be removed in Diffusers v1.0.0. We
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We

```

pipe = StableDiffusionPipeline.from_pretrained(
    "runwayml/stable-diffusion-v1-5"
)
pipe = pipe.to("cuda")

```

```

Loading pipeline components...: 100%              7/7 [00:02<00:00, 2.41it/s]

```

```

Loading weights: 100%              196/196 [00:00<00:00, 636.17it/s, Materializing param=text_model.final_layer_norm.weight]

```

```

CLIPTextModel LOAD REPORT from: /root/.cache/huggingface/hub/models--runwayml/stable-diffusion-v1-5
Key | Status |
-----+-----+
text_model.embeddings.position_ids | UNEXPECTED |

```

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture

```

Loading weights: 100%              396/396 [00:00<00:00, 555.74it/s, Materializing param=visual_projection.weight]

```

```

StableDiffusionSafetyChecker LOAD REPORT from: /root/.cache/huggingface/hub/models--runwayml/stable-diffusion-v1-5
Key | Status |
-----+-----+
vision_model.vision_model.embeddings.position_ids | UNEXPECTED |

```

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture

```

pipe = StableDiffusionPipeline.from_pretrained(
    "runwayml/stable-diffusion-v1-5"
)
pipe = pipe.to("cuda")

```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your setti
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to ac
warnings.warn(

```

```

model_index.json: 100%                               541/541 [00:00<00:00, 19.0kB/
s]

```

```

Download complete: 100%      5.48G/5.48G [03:49<00:00, 72.6MB/s]

```

```

Fetching 15 files: 100%                               15/15 [01:11<00:00, 5.61s/
it]

```

```

Warning: You are sending unauthenticated requests to the HF Hub. Please
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenti

```

```

Loading pipeline components...: 100%                   7/7 [00:03<00:00, 1.65it/
s]

```

```

Loading weights: 100%                               196/196 [00:00<00:00, 417.90it/
s, Materializing param=text_model.final_layer_norm.weight]

```

```

CLIPTextModel LOAD REPORT from: /root/.cache/huggingface/hub/models--runwayml/stable-diffusion-v1-5

```

```

Key                                     | Status      | |
-----+-----+-----+
text_model.embeddings.position_ids | UNEXPECTED | |

```

Notes:

```

- UNEXPECTED :can be ignored when loading from different task/architect

```

```

Loading weights: 100%                               396/396 [00:00<00:00, 583.56it/

```

```

s, Materializing param=visual_projection.weight]

```

```

StableDiffusionSafetyChecker LOAD REPORT from: /root/.cache/huggingface

```

```

Key                                     | Status      | |

```

```
-----+-----+---+
vision_model.vision_model.embeddings.position_ids | UNEXPECTED | |
```

Notes:

- UNEXPECTED :can be ignored when loading from different task/architec

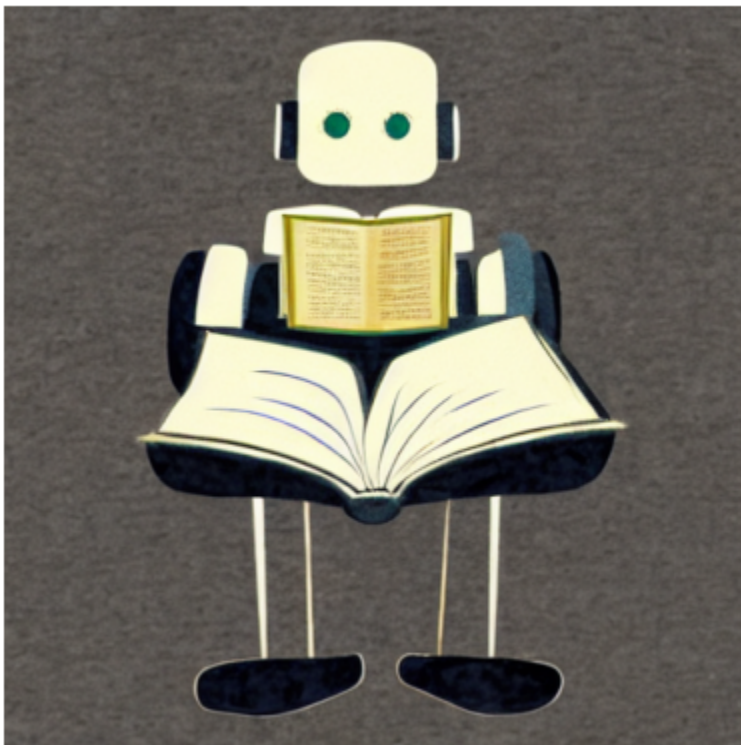
```
prompts = [
    "A futuristic city at sunset",
    "A robot reading a book",
    "A flying car in space",
    "A cyberpunk street",
    "A beautiful mountain landscape"
]
```

```
for i, prompt in enumerate(prompts):
    image = pipe(prompt).images[0]
    image.save(f"image_{i+1}.png")
    print(f"Saved image_{i+1}.png")
```

100%	50/50 [00:23<00:00, 2.19it/s]
Saved image_1.png	
100%	50/50 [00:23<00:00, 2.09it/s]
Saved image_2.png	
100%	50/50 [00:25<00:00, 1.96it/s]
Saved image_3.png	
100%	50/50 [00:27<00:00, 1.82it/s]
Saved image_4.png	
100%	50/50 [00:29<00:00, 1.65it/s]
Saved image_5.png	

```
from PIL import Image
import matplotlib.pyplot as plt
```

```
for i in range(1, 6):  
    img = Image.open(f"image_{i}.png")  
    plt.imshow(img)  
    plt.axis("off")  
    plt.show()
```





```
import os

for file in os.listdir():
    if file.endswith(".png"):
        print(file)
```

```
image_5.png
image_2.png
image_1.png
image_4.png
image_3.png
```

